Effect of Protocol Manipulation for Determining Maximal Aerobic Power on a Treadmill and Cycle Ergometer: A Brief Review

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ABSTRACT

THE DEVELOPMENT OF AEROBIC FITNESS IS CONSIDERED IMPORTANT FOR HEALTH AND PERFORMANCE. IN THIS SENSE, EVALUATION OF MAXIMAL AEROBIC POWER (MAP) BECOMES RELEVANT, NOT ONLY TO CLASSIFY AEROBIC FITNESS, BUT ALSO TO OBTAIN IMPORTANT INDICES RELATED TO MONITORING AND PRESCRIPTION OF EXERCISE, ESPECIALLY HIGH-INTENSITY INTERMITTENT EXERCISE. HOWEVER, IN THE LITERATURE, THERE ARE A VARIETY OF PROTOCOLS THAT MANIPULATE DIFFERENT VARIABLES, SUCH AS INITIAL LOAD, STAGE DURATION, INCREMENT LOAD, AND INCREMENTS BASED ON RATING OF PERCEIVED EFFORT. THEREFORE, THE AIMS OF THE PRESENT PAPER ARE TO ANALYZE STUDIES THAT MANIPULATE THE TEST PROTOCOLS AND THE EFFECTS ON INDICES RELATED TO MAP.

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INTRODUCTION

Maximal aerobic power (MAP) can be defined as the maximal rate at which oxygen can be used by the whole body during severe exercise (4). The assessment of MAP is an important tool for different populations as it allows access to variables related to the state of training, and indices for prescription and monitoring of aerobic exercise. Furthermore, variables related to aerobic fitness can also be used as performance predictors (7). Aerobic fitness is considered an important component for the improvement and maintenance of health, and levels are strongly related to the risk of developing chronic diseases such as type 2 diabetes, hypertension, cardiovascular disease, and dyslipidemia (10).

For the evaluation of MAP, it is common to use graded exercise tests (GXTs), as they are the most appropriate tests when it is necessary to obtain maximal levels of the variables related to aerobic fitness (39). Reproducibility of the maximal mechanical and physiological parameters in GXTs is high on both the cycle ergometer (48) and treadmill (32). Roffey et al. (48) evaluated the reproducibility of maximum parameters (MAP and maximal oxygen consumption, $\dot{V}O_2$max) in 3 different GXTs (SS: stages of 1 minute; LS: stages of 3 minutes; CL + SS: constant load in first stage for 4 minutes following stages of 1 minute). The values obtained in 2 trials were similar for both parameters (MAP and $\dot{V}O_2$max; $P > 0.05$), with the exception of LS GXTs because the values in trial 2 were lower compared with trial 1. For the treadmill, Lourenço et al. (32) repeated the same GXT (3 minutes at 8–8.5 $km \cdot h^{-1}$, plus stages of 25 seconds, with increments of 0.3 $km \cdot h^{-1}$) 4 times and analyzed the reproducibility including coefficient of variation (CV) and typical error (TE) of relative $\dot{V}O_2$max and MAP. No differences were observed in either parameter for the 4 trials

KEY WORDS:
exercise; oxygen consumption; physical fitness

($P > 0.05$). The $\dot{V}O_2$max had a CV of 8.5% and a TE of 4.27 $mL \cdot kg^{-1} \cdot min^{-1}$ (95% confidence interval = 3.23–5.64 $mL \cdot kg^{-1} \cdot min^{-1}$), and the MAP had a CV of 2.3% and a TE of 0.43 $km \cdot h^{-1}$ (95% confidence interval = 0.33–0.57 $mL \cdot kg^{-1} \cdot min^{-1}$).

Although it is generally agreed that GXTs are the best way to assess these levels of aerobic fitness, there are some protocol variables (initial load, duration of stages, load increment, total duration of the test, and even the freedom to self-select the increment of effort) that, when manipulated, can affect the responses obtained from the test (3,6,8,11,31,34,35,48,50,52,53,56). Thus, it is of interest to the evaluator that the choice of protocol for estimating MAP components is the one that results in the best estimate of these components, and that this choice does not modify the values obtained. Thus, the initial choice for a protocol to be adopted is a very important step in the evaluation of the components of MAP.

For experienced evaluators, the choice should be facilitated because experience makes it possible to establish the best combination of manipulated variables in the GXT. However, for evaluators with little experience this choice is not so easy. It would be expected, therefore, that there would be some guidelines in the literature to assist in the choice of the most appropriate protocol. However, the main guidelines provide general recommendations that contain a wide variety of protocols. Thus, the purpose of this review is to analyze the manipulated variables of GXTs and their possible influence in obtaining the variables related to MAP on a treadmill and cycle ergometer, trying to establish some directions for choosing a protocol.

METHODS

The search for articles presented in this review was conducted from databases (PubMed and Web of Science) in 2016. As a search strategy to identify articles, the following keywords were used: "incremental test," "maximal oxygen uptake," "maximal aerobic power," "maximal aerobic speed," "maximum oxygen uptake," "maximum aerobic power," "maximum aerobic speed," "aerobic fitness," and "graded exercise test" without setting a date limit. After this search, the researchers read the summaries to check consistency with the proposal of the review and selected articles that compose this review.

DISCUSSION

ASSESSMENT OF AEROBIC POWER

MAP is estimated by the $\dot{V}O_2$max that can be defined as the maximal rate at which oxygen can be captured and used during severe exercise (4). $\dot{V}O_2$ responses are represented by the linear increase in $\dot{V}O_2$ because of increments in the load (speed for running or power for the cycle ergometer) until the subject reaches $\dot{V}O_2$max, after which a plateau occurs regardless of the increase in load (46). However, not all subjects undergoing such protocols present this plateau value (21) in the $\dot{V}O_2$max and when this plateau does not occur the maximum oxygen uptake is called $\dot{V}O_2$peak.

There is also interest in maximal mechanical response (power or speed) attained during the test, which is used both for monitoring and prescription of aerobic training; although the values of $\dot{V}O_2$max may be stable in highly trained people, improvements in mechanical indices can be noted (15). Moreover, the identification of mechanical load or intensity related to MAP could be a useful tool to prescribe exercise training in practical terms as few strength and conditioning professionals have limited daily access to equipment to measure $\dot{V}O_2$, mainly in high-intensity exercise. Identification of this intensity can guide the prescription of high-intensity intermittent exercise, given that the current literature presents physiological responses important to the prescription, considering the MAP (13).

In general, the tests to evaluate MAP are GXTs divided into stages to increase the load/speed. The increase can be applied in 2 ways: (a) continuous increases in intensity throughout the test (e.g., 6 W every 10 seconds); (b) load increments followed by time (e.g., 36 W per minute). There is no clear definition in the literature regarding the nomenclature for these 2 types of tests; however in this review, protocols with lower increments will be considered as ramp protocols and protocols with higher increments will be referred to as graduated protocols.

In the studies reviewed, it was observed that the authors (30,37,39,52) do not isolate a single variable to be manipulated in the MAP protocols. This form of analysis makes it difficult to understand the influence of each variable, and it can only be inferred if the interference in the test parameters is the result of the sum of the variables or not. In addition, different studies (22,30,56) have subjects with different characteristics regarding the level of training and sex. Accordingly, there is also a limitation in understanding as to whether the various findings result from manipulation or due to the different characteristics of the participants. In addition, some studies performed different protocols on the same day (19), whereas others used protocols that rely on performing a pretest (6). These limitations also limit the advancement of knowledge as to which protocol is most suitable for the parameters that the evaluator seeks to determine, and which is suitable for the population to be evaluated.

CYCLE ERGOMETER GRADED EXERCISE TESTS

According to the literature, the cycle ergometer GXT starts with a lower load (50–120 W) that increases by stages, with a duration varying between 1 and 5 minutes. The load increments at each stage can vary from 25 to 64 W and some authors even use self-selected increases from the rating of perceived exertion (RPE) (3,16,22,35,53) and others reported cadence control (60–90 rpm) (54,55). Some studies (55,57) reported cadence control (60–90 rpm), whereas others allowed participants to choose the cadence (18). Some studies have shown that manipulation of the duration and stage increments can influence the total duration of the test and the responses associated with aerobic fitness (mechanical and physiological). Table 1 presents the main studies that aimed to manipulate the stage duration (6,8,31,48) and load increments (11,50,56) to assess MAP.

One aspect to note is that protocols with longer stages involve tests with longer durations. There are some recommendations (2,23) that to achieve $\dot{V}O_2$max, the test requires a total duration of between 6 and 12 minutes. This paradigm was created from the suggestions made in the study by Buchfuhrer et al. (11), in which this duration variation found the highest values of $\dot{V}O_2$max. However, the study was conducted with only 5 physically active men and did not aim to evaluate the duration of the stages because the manipulated variable was the load increment, a topic that will be discussed later.

Therefore, studies (6,8,31,48) conducted to manipulate the duration of the stages (1, 3, and 5 minutes) indicate that regardless of the choice of the duration of the protocol stages, the $\dot{V}O_2$max/$\dot{V}O_2$peak on the cycle ergometer is not affected. In addition, the total duration of these tests ranged from 10 to 26 minutes with no difference in $\dot{V}O_2$max/$\dot{V}O_2$peak, indicating that long tests are effective to achieve $\dot{V}O_2$max. Migdley et al. (38) suggest that for valid assessment of $\dot{V}O_2$max/$\dot{V}O_2$peak, cycle ergometer GXTs should last between 7 and 26 minutes.

Some studies (11,56) demonstrated significant differences in $\dot{V}O_2$max/$\dot{V}O_2$peak values by manipulating the load increment. These studies showed that in protocols with lower load increments, the achieved values were lower when compared with larger load increments. Buchfuhrer et al. (11) evaluated the $\dot{V}O_2$max kinetics with 3 different increments (10, 30, and 50 W) and found higher values of $\dot{V}O_2$max with intermediate increments (30 W). In turn, Weston et al. (56) found higher values of $\dot{V}O_2$peak in protocols with increases of 30 and 50 W when compared with protocols of 10 W. The difference in the findings of Weston et al. (56) and Buchfuhrer et al. (11) for increments of 50 W may be due to the training level of participants because the sample in the study by Weston et al. (56) was composed of well-trained individuals, which may have allowed a higher increase in the load, unlike the sample of Buchfuhrer et al. (11) which comprised physically active individuals.

Buchfuhrer et al. (11) attributed these differences in $\dot{V}O_2$max to total test duration, being higher for protocols with smaller increases, resulting in higher elevation of body temperature, dehydration, respiratory muscle fatigue, and differences in the use of energy substrates. Weston et al. (56) used a similar justification, indicating that dehydration can lead to an increase in temperature, which in turn would lead to redistribution of the central blood volume, reducing cardiac output. Corroborating these assumptions, Gordon et al. (24) compared 2 conditions: before and after blood donation ($\sim$450 cm^3) in $\dot{V}O_2$max, and observed a reduction in this physiological variable after donating blood (51.3 $\pm$ 7.6 to 48.4 $\pm$ 7.9 mL$\cdot$kg$^{-1}\cdot$min^{-1}). However, the results are still contradictory, as the physiological changes in response to the total duration of the test, as mentioned above in the study by Buchfuhrer et al. (11), were higher in protocols with lower load increments. Furthermore, these explanations are not based on studies using GXTs. If these reasons were entirely true, studies that manipulated the stage duration and had the longest test durations would possibly have observed reductions in$\dot{V}O_2$max/$\dot{V}O_2$peak, which in fact, did not occur. This demonstrates that further studies are needed to assess the relationship between these physiological responses and $\dot{V}O_2$max/$\dot{V}O_2$peak. Furthermore, Scheuermann et al. (50) found no difference in $\dot{V}O_2$max/$\dot{V}O_2$peak when comparing protocols with increases of 8 and 64 W per minute.

The protocols with lower load increments also resulted in lower MAP values (mechanical indices) (11,56). As is seen for $\dot{V}O_2$max/$\dot{V}O_2$peak, there is no explanation for this finding and this does not reflect differences in total work done, which could be a justification for the findings, because, as previously seen, different protocols may result in different values of accumulated work. Analysis of this variable would be useful to justify the observed differences in MAP. Zhang et al. (57) evaluated the effects of 4 protocols that equalized the work performed: (a) ramp protocol with 30 W$\cdot$min^{-1}; (b) graduated 30 W$\cdot$min^{-1}; (c) graduated with 60 W$\cdot$2 min^{-1}; (d) graduated with 90 W$\cdot$3 min^{-1}. None of the evaluated variables were different between protocols, demonstrating the importance of equalization of external work rate generated for comparison of different protocols.

Scheuermann et al. (50), in addition to analyzing the response of $\dot{V}O_2$max/$\dot{V}O_2$peak and MAP, analyzed the root mean square (RMS) of the lateral and medial vastus muscles. This measure represents the recruitment of motor units and is obtained through electromyography (43). The highest values of MAP were observed in the 64 W protocol, which were accompanied by higher RMS values (% of maximal voluntary contraction), and the relationship between the work and RMS was equal between the 2 protocols. As at the end of the test the $\dot{V}O_2$max value was equal between protocols, consequently the $\dot{V}O_2$ and performed work ratio, and $\dot{V}O_2$peak and RMS ratio were higher in the 8 W protocol, demonstrating an increase in mechanical efficiency because with the same $\dot{V}O_2$ value in the 64 W protocol, participants reached higher values of MAP.

As in manipulation of the stage duration, the choice of the load increment is an important variable for defining the MAP reached (50,56). According to the studies, larger increments and, consequently, shorter durations allow higher loads to be achieved.

Although many tests have been conducted to determine $\dot{V}O_2$max, some authors question these protocols indicating that these values cannot be the maximal (35,42,51). Noakes (42) suggested that the way in which some protocols are performed can fail to assess the true value

Table 1
Studies that aimed to evaluate the effects of stage duration and load increment on variables related to aerobic fitness in graded exercise tests on a cycle ergometer

Authors	Sample	Protocols	$\dot{V}O_2$max/$\dot{V}O_2$peak ($L \cdot min^{-1}$)	MAP (W)	TD (min)	Results
Stage duration						
			$mL \cdot kg^{-1} \cdot min^{-1}$			
Bishop et al. (8)	W: 8 PA	$50\ W + 25\ W \cdot min^{-1}$ $50\ W + 25\ W \cdot 3\ min^{-1}$	42.20 ± 6.05 43.70 ± 4.56	284.4 ± 46.2 250.0 ± 37.8	10 ± 2 26 ± 4	$\dot{V}O_2$peak: NS MAP: 1 > 2 TD: NR
			$mL \cdot kg^{-1} \cdot min^{-1}$			
Bentley et al. (6)	M: 9 TRI	$<150\ W + 30\ W \cdot min^{-1}$ $50\%\ MAP + 5\% \cdot 3\ min^{-1}$	62.8 ± 4.7 61.6 ± 5.8	424.0 ± 24.6 355.0 ± 16.7	UN	$\dot{V}O_2$peak: NS MAP: 1 > 2
			$mL \cdot kg^{-1} \cdot min^{-1}$			
Roffey et al. (48)	M: 10 PA	$50\ W + 30\ W \cdot min^{-1}$ $50\ W + 30\ W \cdot 3\ min^{-1}$ $0\ W + 30\ W \cdot 4\ min + 30\ W \cdot min$	50.9 ± 5.5 50.6 ± 4.5 50.3 ± 4.6	329.0 ± 46.8 270.3 ± 41.2 332.0 ± 48.8	10 ± 2 25 ± 4 13 ± 2	$\dot{V}O_2$max: NS MAP: 2 < 1 and 3 TD: 1 < 3 < 2
Lemos et al. (31)	M: 9 PA	$15\ W \cdot min^{-1}$ $50\ W \cdot 3\ min^{-1}$ $50\ W \cdot 5\ min^{-1}$	2.68 ± 1.0 2.58 ± 1.0 2.99 ± 1.3	183 ± 57 153 ± 29 182 ± 43	11 ± 1 10 ± 3 24 ± 3	$\dot{V}O_2$max: NS MAP: NS TD: 3 > 2 e 1
Load increment						
Buchfuhrer et al. (11)	M: 5 PA	$0 + 15\ W \cdot min^{-1}$ $0 + 30\ W \cdot min^{-1}$ $0 + 60\ W \cdot min^{-1}$	3.35 ± 0.38 3.77 ± 0.43 3.62 ± 0.41	NR	18 ± 2 11 ± 1 6 ± 0	$\dot{V}O_2$max: 1 < 2 and 3; 2 > 3 TD: NR
Scheuermann et al. (50)	M: 7 PA W: 2 PA	$0 + 8\ W \cdot min^{-1}$ $0 + 64\ W \cdot min^{-1}$	3.36 ± 0.22 3.31 ± 0.26	252 ± 17 359 ± 24	UN	$\dot{V}O_2$peak: NS MAP: 1 < 2
Weston et al. (56)	M: 20 CYC e TRI	$75\ W + 10\ W \cdot min^{-1}$ $75\ W + 30\ W \cdot min^{-1}$ $75\ W + 50\ W \cdot min^{-1}$	4.65 ± 0.53 4.89 ± 0.56 4.88 ± 0.57	354 ± 38 409 ± 42 437 ± 38	28 ± 4 11 ± 1 7 ± 1	$\dot{V}O_2$peak: 1 < 2 e 3 MAP: 1 < 2 < 3 TD: 1 > 2 > 3

CYC = cyclists; M = men; MAP = maximal aerobic power; NR = values not reported; NS = no significant difference; PA = physically active; TD = test duration; UN = values unvalued; $\dot{V}O_2$max = maximal oxygen consumption; $\dot{V}O_2$peak = peak oxygen uptake; W = women; TRI = triathletes.

of $\dot{V}O_2max$ because they do not correspond with the way the exercises are practiced by the population (e.g., the subjects do not know when the end of the test is, the intensity increases progressively until they cannot continue, and the subjects cannot choose the intensity at which they are exercising) and psychological factors may influence the maximal voluntary exhaustion decision (42). Based on these arguments, some studies have evaluated the influence of autonomy to establish their own pace when carrying out a maximal incremental test (3,16,22,35,53) (Table 2).

Mauger and Sculthorpe (35) compared the effects of a ramp test, during which increases were self-selected by the subjects and based on the RPE, with a traditional protocol to assess $\dot{V}O_2max$. The results showed that the group with the self-selected pace presented higher values of $\dot{V}O_2max$ (40 ± 10 $mL \cdot kg^{-1} \cdot min^{-1}$ versus 37 ± 8 $mL \cdot kg^{-1} \cdot min^{-1}$) and MAP (273 ± 58 W versus 238 ± 55 W) compared with the traditional protocol. However, this study received some criticism (17), and the main one being related to the difference in total duration of the traditional protocol (30% longer, ~13 minutes) compared with the self-selected pace test (10 minutes).

From these findings, Chidnok et al. (16) aimed to compare different protocols equalized for total test duration. In the first protocol, the increments were 30 $W \cdot min^{-1}$ and the rhythm was self-selected. The total duration of this protocol was used to equalize the remaining protocols: (a) ramp with self-selected pace, maintained constant throughout the test; (b) ramp with increases of 30 $W \cdot min^{-1}$ and self-selected pace, varying according to the preference of the evaluated; (c) self-selected protocol based on the RPE–7 stages lasting x seconds (x = ramp test duration divided by 7), and variance in load increments based on RPE. The authors found no differences in $\dot{V}O_2max$ between the different protocols equalized with the same total duration. These findings do not corroborate the results of Mauger and Sculthorpe (35), demonstrating that the total duration of the test should be controlled. It is worth noting that there are limitations in the study by Chidnok et al. (16), because only 7 participants were evaluated and the protocols allowed the participants to choose their own pace, not reproducing the comparison performed by Mauger and Sculthorpe (35).

Subsequently, other studies have been conducted to understand the influence of self-selected pace in tests to evaluate MAP (3,22,53). Straub et al. (53) found no differences in $\dot{V}O_2max$ when comparing a traditional protocol and self-selected protocol based on the RPE (identical to the protocol used by Chidnok (16) with a traditional protocol for evaluation of $\dot{V}O_2max$). In an additional analysis comparing the same participants, considering the preferred protocol, the authors (53) observed higher values of $\dot{V}O_2max$ in the preferred protocols, supporting the assertion that psychological factors modify $\dot{V}O_2max$. However, the findings for the self-selected pace are still controversial because the values of $\dot{V}O_2max$ may be higher (3 or not) (22).

Regarding MAP, many studies have shown that this variable is affected, as in protocols with greater total test duration the MAP reached is lower. The fact that in protocols with shorter stages, individuals can achieve higher values of power, with lower work and energy costs may indicate a lower level of exhaustion, as the accumulated work is different. For example, Amann et al. (1) investigated the effects of 2 GXTs (short and long protocols) in 15 well-trained cyclists. The MAP values were higher for the short protocol (402 ± 35 W, time of ~16 minutes) when compared with the long protocol (363 ± 29 W, time of ~19 minutes). Thus, when the athletes were pedaling at the load of 300 W in the long protocol, they pedaled for 15 minutes and in the short protocol, 12 minutes. This means that at the power of 300 W, in the short protocol, participants performed 18 kJ, whereas in the long protocol participants performed 54 kJ. Another study that demonstrated that in shorter protocols the external work is smaller was conducted by Peiffer (44). During protocols with 1-minute stages, participants performed lower total work compared with the 3-minute protocol; however, the authors did not report the total duration of the test.

Another variable that can be manipulated in cycle ergometer GXTs is pedal cadence. Regarding the effect of pedal cadence on MAP and $\dot{V}O_2max$/$\dot{V}O_2peak$ parameters, we found few studies that investigated such variables (5,20,58). These studies have similarities such as stage duration (3 minutes) and physically active subjects (except Beneke et al. (5) that used triathletes compared with physically active subjects), but had some differences such as in load increments of 30 W (58), 33 W (20), and 50 W (5). Concerning the results obtained for these studies, we can observe that the $\dot{V}O_2max$/$\dot{V}O_2peak$ was not affected by pedal cadence. Denadai et al. (20) did not measure $\dot{V}O_2max$/$\dot{V}O_2peak$ in their study.

Conversely, the MAP seems to have been affected by pedal cadence in 3 studies (5,20,58). In Zoladz et al. (58), study cadences at 60, 80, and 100 revolutions per minute resulted in the same MAP, but at 40 and 120 revolutions per minute the MAP was reduced. Denadai et al. (20) reported that MAP was 8% less at 100 revolutions per minute when compared with 50 revolutions per minute; finally, Beneke et al. (5) observed that MAP was 7.8% higher at 105 revolutions per minute than 60 revolutions per minute independent of training status (physically active or triathletes).

The MAP in self-selected pace protocols has also been evaluated. Evans et al. (22) compared a self-selected pace protocol with a traditional protocol and observed that the MAP was higher, and the time to exhaustion lower in a self-selected pace protocol. These findings corroborate the literature findings that demonstrate that in shorter protocols the MAP is maximized without altering $\dot{V}O_2$max.

An aspect to be considered in cycle ergometer GXTs is the body position (seated or standing). In last stages of GXTs it is common that the evaluated tend to assume a standing position to extend the test and reach greater outcomes. Some studies investigated the effect of seated or standing in the final stages of cycle ergometer GXTs on $\dot{V}O_2$max values (9,29,40,55) and most of them demonstrated that the $\dot{V}O_2$max values were similar in both position (29,40,55). It is key to highligth that the MAP was not investigated. Thus, the $\dot{V}O_2$max seems to not be affected by body position and this strategy appear to be unnecessary to improve this response.

Table 3 presents the summary of information about manipulated variables of cycle ergometer GXTs.

Table 2
Studies that aimed to evaluate the effects of self-paced and rating of perceived exertion on variables related to aerobic fitness in graded cycle ergometer tests

Authors	Sample	Protocols	$\dot{V}O_2$max (mL·kg^{-1}·min^{-1})	MAP (W)	TD (min)	Results
Mauger and Sculthorpe (35)	M: 16 NT	60 W + 30 W·2 min^{-1} 5 × 2-min; I = Borg–11, 13, 15, 17, 20	37 ± 8 40 ± 10	238 ± 55 273 ± 58	13 ± 3 10 ± 0	$\dot{V}O_2$max: 1 < 2 MAP: 1 < 2 TD: 1 > 2
Chidnok et al. (16)	M: 7 PA	0 W + 30 W·min^{-1} (steady pace) 0 W + 30 W·min^{-1} (free pace) 7 × "X" s* + I = Borg–8, 10, 12, 14, 16, 18, 20	~57.73 ~57.46 ~58.13	UN	13 ± 2 13 ± 2 13 ± 2	$\dot{V}O_2$max: NS TD: NS
Straub et al. (53)	M: 20 CYC W: 4 CYC	M: 80 + 30 W·min^{-1} W: 80 + 20 W·min^{-1} 5 × 2-min; I = Borg–11, 13, 15, 17, 20	~51.88 ~52.02	330 ± 71 317 ± 97	10 ± 1 10 ± 0	$\dot{V}O_2$max: NS MAP: NS TD: NS
Evans et al. (22)	M: 10 PA W: 6 PA	M: 20 W + 13 W·min^{-1}; W: 20 W + 10 W·min^{-1} 3 min^{-1} with I = Borg–11, 13, 15, 17, 20	44.3 ± 4.86 43.5 ± 4.05	258 ± 54 279 ± 70	21 ± 2 17 ± 1	$\dot{V}O_2$max: NS MAP: 1 < 2 TD: 1 > 2
Astorino et al. (3)	M: 19 PA W: 11 PA	M60– 80 W + 30–40 W·min^{-1} W50–60 W + 25–30 W·min^{-1} 5 × 2-min; I = Borg–11, 13, 15, 17, 20	47.2 ± 10.2 50.2 ± 9.6	306 ± 70 307 ± 95	10 ± 1 10 ± 1	$\dot{V}O_2$max: 1 < 2 MAP: NS TD: 1 > 2

* = Duration of the stages were calculated from total test time divided by 7; ~ = Absolute values converted to relative; CYC = cyclists; I = increments; M = men; MAP = maximal aerobic power; NS = no significant difference; NT = not trained; PA = physically active; TD = test duration; UN = values unvalued; $\dot{V}O_2$max = maximal oxygen consumption; W = women.

TREADMILL GRADED EXERCISE TESTS

Treadmill GXTs start with a low intensity which is progressively increased until volitional exhaustion. In the literature, there are a variety of protocols that manipulate certain variables, such as initial speed, speed increment, stage duration, inclusion of pauses between stages, and freedom to choose self-selected pace. Some of the studies (27,28,30,33,37,39,45) that aimed to compare different progressive treadmill test protocols are presented in Table 4.

For initial load, a difference can be observed only in the initial speed (37), and speed and slope together (27). In initial speed, a variation between 2.7 and 14.4 km·h^{-1} was observed (27,45) and for initial slope, the variation was between 0 and 10% (27,33). The increments in speed ranged between 0.5 and 2.1 km·h^{-1} (39,45) or through individual choice (26,34,52). In turn, slope increments ranged between 1 and 5% (45). Another variable that has been manipulated is stage duration–between 1 and 6 minutes (30). There are also

Table 3
Summary of information about manipulated variables of graded cycle ergometer tests

	Influence on maximal aerobic power or $\dot{V}O_2max/\dot{V}O_2peak$
Load increment	Small increments (lower than 30 $W \cdot min^{-1}$ for sedentary, and 30/50 $W \cdot min^{-1}$ for trained individuals) negatively affect the maximal aerobic power and $\dot{V}O_2max/\dot{V}O_2peak$ values.
Cadence	Lower (~40 rpm) or higher cadences (~120 rpm) negatively affect the maximal aerobic power, but do not affect $\dot{V}O_2max/\dot{V}O_2peak$ values.
Stage duration	Protocols with shorter duration (1 min) allow higher values to be obtained for maximal aerobic power, but do not affect $\dot{V}O_2max/\dot{V}O_2peak$ values.

GTXs = graded exercise tests; $\dot{V}O_2max$ = maximum oxygen uptake; $\dot{V}O_2peak$ = peak oxygen uptake.

studies that included a pause of 30 seconds between stages (39). Thereafter, to manipulate certain variables of the testing protocol, a modification was observed in the total test duration, which implies manipulation of the increment to ensure the evaluated individual presented exhaustion at the established time (28).

Manipulation of treadmill slope as an intensity increase in GXTs is understandable. However, the application of mechanical loads obtained in these tests for the prescription of aerobic training is limited because the evaluated individuals will not always have access to a treadmill with an incline to train on. In addition, when obtaining the maximal mechanical load associated with high inclinations, the participant will find it impossible to run at high speeds. This occurrence could be an implication of some protocols used by McConel and Clark (37) and Kang et al. (27), as at the end of the test, the evaluated individual could reach maximal slopes of between 20 and 25%; however, the recommendations suggest that the slopes do not exceed 15% (38). Some limitations can also be found for the application of mechanical loads obtained in GXTs, including pauses between stages, for example, because this pause can influence the MAP (39).

An indication in the literature is related to total duration of GXTs (between 8 and 12 minutes) and therefore, any combination that allows the evaluated individual to reach exhaustion within this total duration is an appropriate protocol. This paradigm is based on the study by Buchfuher et al. (11), and its implications were outlined in the previous section, demonstrating that this statement is not supported and is very limited (38).

It is possible that manipulation of the protocols results in differences when obtaining the physiological and mechanical indices related to MAP. However, this should not occur, and if confirmed has implications for the use of the parameters obtained with GXTs for the practical application of the results and comparison of the studies with research applications.

The studies presented in Table 4 present great variation in the protocols used, the samples evaluated, and the results obtained. For the different protocols, it is observed that the manipulation of more than 1 variable in the same study and the different characteristics of the training level (sedentary, physically active, and trained) and sex (male and female) make it difficult to compare results. Consequently, any inference about the effect of a single variable cannot be established, only a combination of this variable with other variables.

The main physiological index compared between studies was $\dot{V}O_2max/\dot{V}O_2peak$ to estimate the MAP, and inconsistent findings were observed regarding the influence of the protocol on the value of $\dot{V}O_2max/\dot{V}O_2peak$; in some studies, this difference was observed (33,45), in others it was not observed (28,30,37,39), and there was even inconsistency between different samples from the same study (27). It is also noted that studies that observed this difference used very different protocols, that is, many variables manipulated between the protocols (27,33,44), which could be an explanation for the findings, as studies that manipulated a lower number of variables observed a difference in $\dot{V}O_2max/\dot{V}O_2peak$ values (28,30,37,39).

It is noteworthy that, even using different protocols, Kang et al. (27) observed that only trained athletes presented differences in $\dot{V}O_2max$, whereas sedentary and physically active individuals were similar regarding values obtained. In turn, Pollock et al. (45), comparing $\dot{V}O_2max$ obtained in different protocols, found that sedentary and physically active individuals had different responses, but these differences were not found between the same protocols. Thus, in cases where obtaining the $\dot{V}O_2max$ value is the purpose of the application of the GXT, the importance of establishing specific guidelines for the implementation of the protocol is reinforced because even if a protocol is reproducible, it is not certain that it offers the best estimate of $\dot{V}O_2max$. In comparison of mechanical response, differences are observed between the protocols (28,30,33,39). Thus, when the interest of application of the GXT is to obtain the maximal mechanical load (MAP), manipulation will result in different responses of this parameter.

Finally, the total duration of the test was compared between the different protocols, and the findings are

Table 4
Studies that aimed to evaluate the effects of the initial speed, speed increment, stage duration, and total duration of the test on variables related to aerobic fitness in graded treadmill tests

Author	Sample	Protocols	$\dot{V}O_2$max/$\dot{V}O_2$peak (mL·kg^{-1}·min^{-1})	MAS (km·h^{-1})	TD (min)	Result
Pollock et al. (45)	M: 29 NT M: 22 PA	Åstrand modified IS: 8–13.7 km·h^{-1} and 0%; I: 2.5%; SD: 2-min Balke—IS: 5.3 km·h^{-1} and 0%; I: 1%; SD: 1-min Bruce—IS: 2.7 km·h^{-1} and 10% SI: 0.8–1.3 km·h^{-1}; I: 2%; SD: 3-min Ellestad—IS: 2.7 km·h^{-1} and 10% SI: 1.6–2.1 km·h^{-1}; I: 0–5%; SD: 2-min	37.7 ± 4.2 35.8 ± 4.1 35.3 ± 3.9 36.3 ± 4.4 47.3 ± 5.4 44.1 ± 4.4 46.3 ± 5.7 46.7 ± 5.5	UN	7.4 ± 1.1 14.7 ± 2.7 9.4 ± 1.1 8.3 ± 0.1 8.3 ± 1.0 19.8 ± 2.9 11.5 ± 1.0 10.5 ± 1.4	$\dot{V}O_2$max: 1 > 3 TD: NC $\dot{V}O_2$max: 1 > 2 TD: NC
McConnel and Clark (37)	M: 10 TR	IS: 12.9 km·h^{-1} and 0%; I: 2.5%; SD: 1-min IS: 12.9 km·h^{-1} and 0%; I: 2.5%; SD: 2-min Training pace IS: 14 ± 0.8 km·h^{-1} and 0%; I: 2.5%; SD: 2-min Selected by athlete IS: 12.7 ± 1.8 km·h^{-1} and 0%; I: 2.5%; SD: 2-min	65.0 ± 5.6 64.5 ± 5.3 66.2 ± 3.9 64.7 ± 5.8	UN	10.1 ± 0.6 13.1 ± 1.2 11.8 ± 1.1 13.6 ± 2.5	$\dot{V}O_2$max: NS TD: 1 < 2, 3 and 4; 3 < 4
Lukaski, Bolonchuk and Klevay (33)	M: 16 PA	Balke—IS: 5.3 km·h^{-1} and 0% I: 2%; SD: 1 min Bruce—IS: 2.7 km·h^{-1} and 10% SI: 0.8–1.3 km·h^{-1}; I: 2%; SD: 3-min Ellestad—IS: 2.7 km·h^{-1} and 10% SI: 1.6–2.1 km·h^{-1}; I: 0–5%; SD: 2-min	 45.6 ± 1.4 47.9 ± 1.5 47.4 ± 6.3	PP Watts$^{-1\#}$ 236 ± 11 260 ± 12 284 ± 11	 27.6 ± 0.7 18.0 ± 0.4 16.4 ± 0.2	 $\dot{V}O_2$peak: 1 < 2 and 3 PP: 1 < 3 TD: 1 > 2 and 3
Kang et al. (27)	M: 15 NT M: 12 TR W: 10 NT	Åstrand—IS: 9.7 km·h^{-1} and 0% I: 2.0%; SD: 2-min Bruce—IS: 2.5 km·h^{-1} and 10% SI: 1.1–1.4 km·h^{-1}; I: 2%; SD: 3-min Costill/Fox—men M—IS: 14.4 km·h^{-1} and 0%; W—IS: 12.6 km·h^{-1} and 0% I: 2.0%; SD: 2-min	VG	UN	9.8 ± 0.5 12.4 ± 0.4 4.9 ± 0.3 14.5 ± 0.5 17.0 ± 0.5 10.4 ± 0.4 9.0 ± 0.8 11.0 ± 0.6 5.3 ± 0.6	$\dot{V}O_2$max: NS $\dot{V}O_2$max: 2 < 1 and 3 $\dot{V}O_2$max: NS
Kuipers et al. (30)	M: 4 TR W: 4 TR	M—IS: 10 km·h^{-1} and 1% W—IS: 8 km·h^{-1} and 1% I: 1%; SD: 1-min; I: 2%; SD: 3-min I: 2%; SD: 6-min	L·min^{-1} 4.48 ± 1.13 4.46 ± 1.04 4.41 ± 1.01	 18 ± 2 17 ± 2 15 ± 2	 UN	 $\dot{V}O_2$peak: NS MAV: 3 < 1 and 2

(continued)

Table 4 (continued)

Midgley et al. (39)	M: 9 TR	Continuous—IS: 9 $km \cdot h^{-1}$ and 1%; SI: 0.5–1 $km \cdot h^{-1}$; SD: 1-min Intermittent—IS: 9 $km \cdot h^{-1}$ and 1%; SI: 1 $km \cdot h^{-1}$; SD: 2-min; P: 30 s Intermittent—SI: 9 $km \cdot h^{-1}$ and 1%; SI: 1 $km \cdot h^{-1}$; SD: 3-min; P: 30 s	$L \cdot min^{-1}$ 4.09 ± 0.54 4.10 ± 0.52 3.98 ± 0.49	16 ± 1 17 ± 1 15 ± 1	10.3 ± 1.7 22.2 ± 2.5 30.0 ± 2.8	$\dot{V}O_2$max: NS MAV: 1 < 2; 1 > 3; 2 > 3 TD: NS
Kirkeberg et al. (28)	M: 12 PA	Short protocol—TD: 8-min IS: 8 $km \cdot h^{-1}$ and 3%; SI: variable; SD: 1-min Average protocol—TD: 10 min IS: 8 $km \cdot h^{-1}$ and 3%; SI: variable; SD: 1-min Long protocol—TD: 14-min IS: 8 $km \cdot h^{-1}$ and 3%; SI: variable; SD: 1-min	48.2 ± 5.3 48.9 ± 5.1 49.1 ± 4.7	13 ± 1 13 ± 10 12 ± 1	9.22 ± 1.21 10.78 ± 1.31 13.06 ± 2.25	$\dot{V}O_2$max: NS MAV: 3 > 1 and 2; 2 > 1 TD: 3 > 1 and 2; 2 > 1

I = increment of slope; IS = initial velocity; M = men; MAS = maximal aerobic speed; NS = not significant difference; NT = not trained; P = pause between stages; PA = physically active; peak power expressed in Watts#: peak power calculated = body weight * distance * sinθ/6.12; SD = stage duration; SI = speed increment; TD = test duration; TR = trained runners; UN = values unvalued; VG = value expressed in graphs; $\dot{V}O_2$max = maximal oxygen consumption; $\dot{V}O_2$peak = peak oxygen uptake; W = women.

inconsistent because some studies observed a difference in the obtained indices (33,37), whereas others did not (39). Kirkeberg et al. (28) observed a difference in total duration of the test, but this variation was the objective of the authors because they determined the duration of the test and calculated the speed increment necessary for the sample to reach exhaustion in this period. The mean interval duration observed in these studies was between 5 and 30 minutes (27,39).

An interesting finding observed by Midgley et al. (39) was that, even using protocols resulting in an average duration of 10, 22, and 30 minutes, no difference was observed in $\dot{V}O_2$max value, although the maximal speed varied. Thus, it is observed that the paradigm that shorter tests (between 8 and 12 minutes) are better indicators to obtain the $\dot{V}O_2$max value (11) may not be true as previously stated, although this statement seems to be true for maximal aerobic speed (33,39). It is noteworthy that the assertion that duration can affect the maximal aerobic speed, but not the $\dot{V}O_2$max value, cannot be asserted, because Lukaski et al. (33) also observed that protocols with shorter duration result in higher $\dot{V}O_2$max/$\dot{V}O_2$peak values.

Self-selected pace protocols that aim to evaluate maximal aerobic speed have also been used (26,34,52) (Table 5).

Mauger et al. (34) observed higher values of $\dot{V}O_2$max in protocols with self-selected pace compared with traditional protocols. The findings of this study were criticized (47), as the authors performed the protocol with self-selected pace on a nonmotorized treadmill and the traditional protocol on a motorized treadmill, which may influence the achievement of this variable. Recently, Morgan et al. (41) compared the physiological and performance responses in 2 GXTs performed on a motorized and nonmotorized treadmill. Higher values were observed on the nonmotorized compared with

Table 5 Studies that aimed to evaluate the effects of self-paced and rating of perceived exertion on variables related to aerobic fitness in graded treadmill tests					
Author	**Sample**	**Protocols**	**$\dot{V}O_2$max ($mL \cdot kg^{-1} \cdot min^{-1}$)**	**TD (min)**	**Result**
Mauger et al. (34)	M: 14 TR	IS: not informed and 1%; SI: 1 $km \cdot h^{-1}$; SD: 2-min RPE—5 stages; SI: Borg–11, 13, 15, 17, 20; SD: 2-min	61.4 ± 7.3 64.4 ± 7.7	10 ± 3 10 ± 0	$\dot{V}O_2$max: 1 < 2 TD: NS
Sperlich et al. (52)	M: 40 TR	IS: 12.9 $km \cdot h^{-1}$ e 0%; I: 2.5%; SD: 1-min IS: 12.9 $km \cdot h^{-1}$ e 0%; I: 2.5%; SD: 2-min Training pace IS: 14 ± 0.8 $km \cdot h^{-1}$ e 0%; I: 2.5%; SD: 2-min Selected by athlete IS: 12.7 ± 1.8 $km \cdot h^{-1}$ e 0%; I: 2.5%; SD: 2-min	65.0 ± 5.6 64.5 ± 5.3 66.2 ± 3.9 64.7 ± 5.8	10 ± 1 13 ± 1 12 ± 1 14 ± 3	$\dot{V}O_2$max: NS TD: 1 < 2, 3 and 4; 3 > 4
Hogg et al. (26)	M: 40 TR	Gradual—IS: uninformed and 3% SI: 1 $km \cdot h^{-1}$; SD: 2 min RPE—5 stages; IS: self-paced (RPE); I: 3% SI: Variation in the first stage and the end of the penultimate stage according to RPE (11,17,20); SD: 2-min RPE running—5 stages; IS: self-selected; I: 3%; SD: 2-min	68.6 ± 6.0 70.6 ± 4.3 67.6 ± 3.6	11 ± 1 10 ± 0 10 ± 0	$\dot{V}O_2$max: 2 > 1 and 3 TD: NS

I = increment; IS = initial test load; M = men; NS = not significant difference; RPE = rating of perceived exertion; SD = stage duration; SI = speed increase; TD = test duration; TR = trained runners; $\dot{V}O_2$máx = maximal oxygen consumption.

motorized treadmill for MAP (~14.8 ± 2.3 versus 12.9 ± 1.4 $km \cdot h^{-1}$) and total duration (725 ± 168 versus 595 ± 109 seconds); however, no difference was observed for $\dot{V}O_2$max/$\dot{V}O_2$peak (48.6 ± 9.2 versus 47.8 ± 8.9 $mL \cdot kg^{-1} \cdot min^{-1}$).

Hogg et al. (26), assuming that athletes have high ability to regulate their pace in the test, compared 2 protocols based on self-selected pace to increase slope or speed, and a traditional protocol. In the results, the authors observed that the $\dot{V}O_2$max values were higher only in the self-selected pace with increases in slope when compared with the traditional protocol, suggesting that self-selected pace protocols with increasing speed are similar to traditional protocols when the sample is composed of athletes.

Sperlich et al. (52) evaluated well-trained men in 5 protocols with the aim of measuring $\dot{V}O_2$max: (a) increase in speed/slope; (b) constant speed and increase in slope; (c) constant slope and increase in speed; (d) slope of 1% and increments in speed; (e) self-selected slope and speed. It was found that the different protocols resulted in similar $\dot{V}O_2$max values. Thus, it seems that the self-selected pace to evaluate the MAP in running presents controversial results, which may result from the different protocols compared in the studies (26,34,52).

Table 6 presents the summary of information about manipulated variables of graded cycle ergometer tests.

IMPORTANT ASPECTS TO CONSIDER IN THE CHOICE OF ERGOMETER TYPE

Although it was not the main focus of our review to compare the variables related to MAP on different ergometers, it is important to point out that higher $\dot{V}O_2$max/$\dot{V}O_2$peak values are found in exercise modes that involve more muscle mass, such as treadmill, rowing, and skiing when compared with arm or cycle ergometer exercise modes, for example (36). However, this response can differ if the person is highly trained in a specific mode of exercise/modality. For example, when comparing $\dot{V}O_2$max values on 2 types of ergometers (treadmill and cycle ergometer) in untrained individuals, cyclists, runners, and triathletes, Caputo and Denadai (14) observed that the treadmill test resulted in higher values than the cycling test in runners and untrained individuals. For triathletes, $\dot{V}O_2$max did not differ between exercise modes, whereas for cyclists the highest $\dot{V}O_2$max values were achieved during the cycling test. Thus, training background is a very important variable to consider when different exercise modes are compared.

The available literature does not allow discussions about the manipulation of variables in GXTs on all ergometers, but it is possible to find protocols to determine MAP and $\dot{V}O_2$max in

Table 6
Summary of information about manipulated variables of graded treadmill tests

	Observations
Combination of speed and slope increment and stage duration	The GXTs manipulated different variables in the same study, so the isolated effect of these variables on maximal aerobic power and $\dot{V}O_2max/\dot{V}O_2peak$ is not clear. When these variables are combined, they result in a different total duration of GXTs. For measurement of maximal aerobic power, the combinations of different variables in GXTs can result in shorter total duration (between 8 and 12 min). Longer total durations in GXTs negatively affect the maximal aerobic power. For $\dot{V}O_2max/\dot{V}O_2peak$, the combinations of different variables do not affect $\dot{V}O_2max/\dot{V}O_2peak$ values.

GTXs = graded exercise tests; $\dot{V}O_2max$ = maximum oxygen uptake; $\dot{V}O_2peak$ = peak oxygen uptake.

different exercise modes: running, cycling (including arm cranking), rowing, skating, skiing, and an elliptical ergometer. Moreover, we can observe growing interest in the development of sport-specific tests. Examples of sport-specific tests are the GXT protocols developed for tennis (25), taekwondo (49), and karate (54). For intermittent modalities that involve effort and pause periods, the 30:15 seconds has been used (12) to reproduce the movement patterns. It is important to consider the validity, reproducibility, and sensibility of these specific GXTs.

CONCLUSIONS AND PRACTICAL APPLICATIONS

According to the information available in the literature, it can be concluded that there are a wide variety of GXTs for the cycle ergometer and treadmill, and the application of these leads to different responses. In the majority of studies, the manipulated variables relate to the initial load, load increases, duration of stages, and the freedom to select the pace (intensity increments). Studies manipulate one or more variables in different ways, making it difficult to infer exactly how these variables can affect the parameters obtained in GXTs.

Generally, the variable that suffers less influence from manipulation of the protocols is $\dot{V}O_2max/\dot{V}O_2peak$, which enables the use of various protocols to obtain this physiological index. The maximal aerobic mechanical loads seem to be the most affected variables. In shorter protocols, shorter stages or larger increments generate higher maximal aerobic load values compared

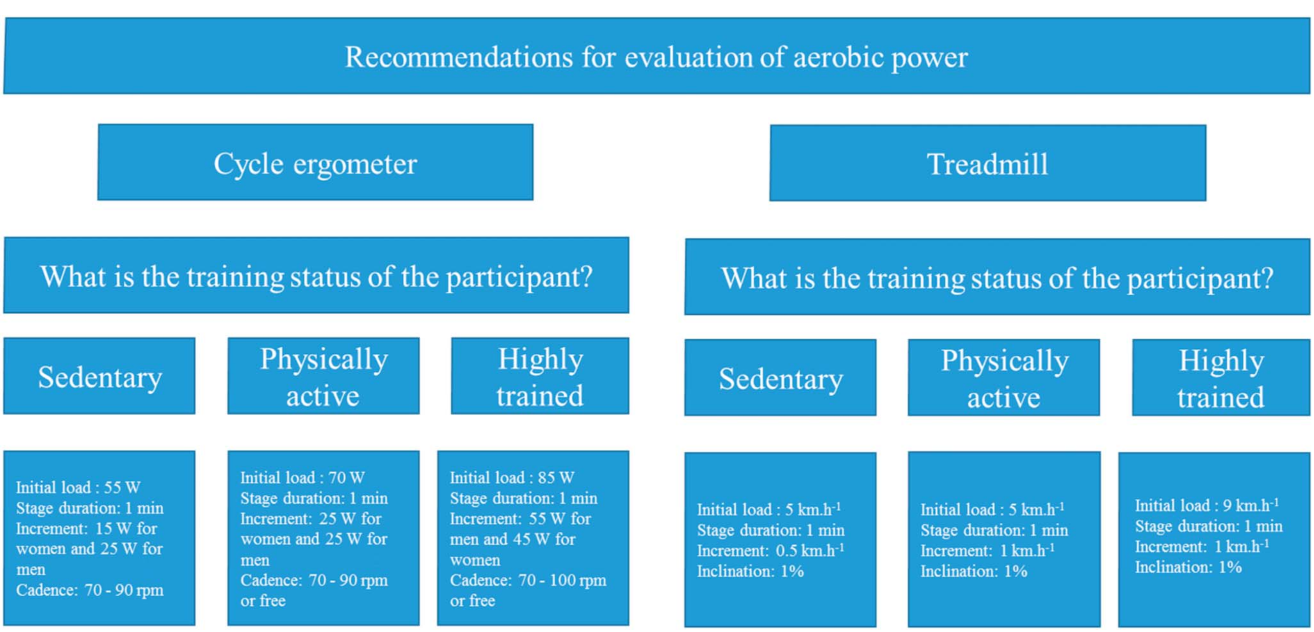

Figure. Suggestions of protocols for cycle ergometer and treadmill graded exercise tests for different populations according to training status.

with longer protocols (longer stages with or lower load increments). Because exercise prescription is often performed according to the intensity that should be maintained and not by constant measurement of a physiological variable, the fact that the mechanical variable is the most affected makes the prescription process fragile, especially when the results of different studies are compared.

Therefore, further studies are needed with more defined methodologies, using a fixed variable (e.g., stage duration or load increase) and exhausting all possibilities for this fixed variable. Together or subsequently, other studies should be performed by manipulating each of the variables in isolation. An important point for elaboration of the protocols for comparison is to equalize the work performed because this variable seems to influence the differences between protocols. In addition, it is necessary to delineate the studies described above in patients with different stages of training and sex, as well as different types of ergometers (treadmill, cycle ergometer, rowing ergometer, elliptical, etc.). The combination of these studies should provide accurate and safe recommendations to evaluators to select the protocol to be used.

We present a flowchart (Figure) with suggestions for cycle ergometer and treadmill GXTs, to assist the strength and condiotining professionals to evaluate the MAP in different populations.

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